View Reviews

Paper ID	516
Paper Title	Adaptive Hybrid Quantum-Classical Neural Network for Medical Diagnosis
Track Name	Track 2: ML in Clinical Decision Support and Predictive Analytics

Reviewer #2

Questions

2. The paper is relevant to the scope of AI in Healthcare and the selected conference track.
Neutral

3. The research methodology is appropriate, technically sound, and adequately explained.
Neutral

4. The experimental results or evaluation adequately support the claims made in the paper.
Disagree

5. The paper discusses relevant prior work and provides appropriate references.
Disagree

6. The paper is clearly written, logically organized, and understandable.
Agree

7. Describe the main strengths and contributions of the paper.
1. Honest scoping. The paper states repeatedly that the workload figures are estimates of layer usage rather than energy or time measurements, that the work is not a demonstration of quantum advantage or clinical readiness, and that SHAP explanations are not causal. It also reports results that do not favour the proposed model (lower CV F1 than the Fixed HQNN on every dataset).

2. Good evaluation hygiene. Imputation and scaling are fit on training folds only, routing thresholds are tuned in a nested inner cross-validation, a holdout test set is kept untouched, and stratified 5-fold CV is used. The treatment of zero-as-missing values in the Pima data is sensible.

3. A useful set of comparisons. Besides a classical MLP and a Fixed HQNN, the authors include an Adaptive Classical Control (M4), which is the right idea for separating the effect of routing from the effect of quantum layers. Three datasets of different dimensionality are used, mean and SD over folds are reported, and paired statistical tests are attempted.

4. Clear architecture. Fig. 1 and Eqs. 1-9 describe the three inference paths clearly, and only the selected path executes at inference. Optimizer, learning rate, batch size, epochs and software (PyTorch, PennyLane default.qubit) are given.

5. Reasonable idea. Bringing conditional computation to hybrid quantum-classical models is a sensible direction given NISQ-era constraints, and the reported layer-usage figures are consistent with the path distributions in Table II.

8. Identify the main weaknesses, limitations, and areas for improvement.

1. No demonstrated benefit from the quantum component. The classical MLP (M1) beats the Adaptive HQNN (M3) in CV on all three datasets, and matches or beats the Fixed HQNN (M2) on Heart and Diabetes (e.g., Heart accuracy 83.45 vs 79.75/76.02). M2's WDBC lead over M1 (98.24 vs 97.80) is within one SD. On WDBC, 98% of samples never reach a quantum layer, so the model is effectively classical. A 4-qubit, 2-layer circuit is also trivially simulable classically. The paper thus economises on a component never shown to help. Missing ablations: a Path-0-only model; a classical block with the same 4-d bottleneck and linear head in place of the circuit; a parameter-matched MLP.

2. The one comparison that isolates the quantum contribution, M3 vs M4, is not analysed. CV differences are 96.92 vs 95.82 (WDBC), 76.02 vs 74.78 (Heart) and 73.94 vs 73.45 (Diabetes), all within one SD, and test F1 is equal or mixed. M4 is not described (what computation is skipped, what is saved), and its routing/workload is absent from Table II, so the claim in Sec. V-E that adaptive classical computation "also yields savings" is unsupported by the data shown. M1's architecture and parameter count are not given either.

3. The statistical testing cannot support its conclusion. With five paired folds, the smallest possible two-sided exact Wilcoxon p-value is 2/2^5 = 0.0625, so $p < 0.05$ is unattainable whatever the effect. The reported p = 0.0625 for Heart and Diabetes means Fixed beat Adaptive in AUC in all five folds. Describing this as "no significant difference detected" with "limited power" is misleading: the test has no power, and the data point to a consistent drop (Heart CV AUC 0.8884 vs 0.8138). Only AUC is tested; F1/accuracy and comparisons against M1 and M4 are not. Repeated CV or multiple seeds with confidence intervals are needed.

4. Holdout results are noisy and over-read. The test sets contain roughly 114, 61 and 154 samples. On Heart, M2, M3 and M4 have identical test F1 (85.25), as do M2 and M4 on WDBC (96.30); one or two samples move F1 by several points. Yet the text says M3 "looks more competitive" on test and that CV results are "confirmed", although CV and test often disagree (Heart M3 F1 73.49 vs 85.25). No CIs are given and only one random seed is used. Table I omits test accuracy, precision and recall, although Sec. IV-F says they are reported. It is unclear which models and thresholds were applied to the holdout (a fold model, an ensemble, or a refit on the full development set).

5. The workload metric is a weak proxy. Eq. 10 assumes cost proportional to layer count, but circuit cost is dominated by state preparation, measurement/shots and queueing, and the classical encoder and preliminary head run for every sample. Path 1 and Path 2 differ by only four rotations and four CNOTs. Neither simulator wall-clock time nor shot/noise simulation is reported, so the NISQ motivation is untested. The percentage reduction mostly reflects how easy each dataset is (WDBC 98.8%), not any quantum-specific property of the routing.

6. Design and training are under-specified, with a possible capacity problem. (a) Paths 1-2 use only a 4-d tanh-bounded projection and a linear 4-to-1 head, whereas Path 0 uses the full 16-d features, so the hardest samples go to the lowest-capacity path. The paper never shows that Paths 1-2 improve on the preliminary prediction for the samples routed there (report preliminary vs final accuracy per path). (b) How the preliminary head, router and both quantum heads are

trained jointly (all paths on all samples? loss weights?) is not stated; quantum layers trained on all samples but used only on hard ones face a distribution shift. (c) The thresholds are called "learned" but are chosen by search under an unstated "predefined objective" and grid; the selected values are not reported, and it is unclear whether inner-fold models or the final model supply the confidences used for selection. (d) Routing uses uncalibrated sigmoid confidence, with no calibration analysis. (e) There is no accuracy-vs-average-layers Pareto curve and no random-routing baseline at equal compute, so confidence-based routing itself is not shown to beat arbitrary routing.

7. Baselines and benchmarks are limited. Only one MLP is used: no logistic regression, SVM, random forest or gradient boosting, which are strong on these small tabular sets, and no comparison with published results (including prior quantum-ML studies on the same datasets). Hyperparameters (lr 0.005, batch 32, 60 epochs) are fixed rather than tuned per model, and early stopping is not described. The datasets are small, saturated and single-source (Pima is a single population), so generalisation is untested.

8. The XAI component is thin. It is a single paragraph with no figures; the model explained, background data and number of samples for Kernel SHAP are not stated. That deeper-path samples sit where the preliminary prediction is less decisive is true by construction, since the router thresholds that confidence. The top features (worst concave points, thalach, Glucose/BMI) are the familiar ones from classical models and say nothing about the quantum component, yet XAI is listed as contribution 5.

9. Related work and clarity. Section II-A has no citations, and there is no discussion of quantum-ML studies on medical/tabular data, adaptive-depth quantum circuits, or classical cascades, selective prediction and learning-to-defer, which are the nearest analogues of confidence routing. The Introduction moves from "A natural way to relax..." to "To address this problem" without ever stating the problem. R(theta) is unspecified (single parameter in Eq. 6, "Rot gates" in Fig. 1), |psi0> is undefined, and Fig. 1 is hard to read. The "estimated"/"not clinical" disclaimers recur many times, while the Heart AUC drop (0.075) is called modest and the low Diabetes F1 (~66%) is not discussed. Relevance to the track is partial: the data are medical, but the contribution is computational with no clinical evaluation.

Reviewer #4

Questions

2. The paper is relevant to the scope of AI in Healthcare and the selected conference track.
Strongly Agree

3. The research methodology is appropriate, technically sound, and adequately explained.
Neutral

4. The experimental results or evaluation adequately support the claims made in the paper.
Neutral

5. The paper discusses relevant prior work and provides appropriate references.
Disagree

6. The paper is clearly written, logically organized, and understandable.
Neutral

7. Describe the main strengths and contributions of the paper.
The paper addresses a relevant research problem and presents a comparative study of multiple algorithms.

The equations and figures strengthen the technical presentation and overall clarity of the paper.

The paper addresses quantum computing, an emerging and rapidly growing research area with significant potential for future technological advancements and practical applications.

8. Identify the main weaknesses, limitations, and areas for improvement.
The citations should be arranged in the order in which they appear in the text, following the conference's prescribed referencing style.
Although the paper mentions comparing four models under an identical data partitioning and evaluation protocol, the results of all four algorithms are not clearly presented or comparatively discussed. A detailed comparison of their performance should be included.
The result analysis needs to be strengthened. The explanations generated by the Explainable AI techniques should be discussed in greater detail, highlighting the important features and their influence on the model predictions.
The references are outdated, with the latest cited reference dating back to 2021. Recent relevant research should be incorporated to establish the novelty and relevance of the proposed work.
In Table 2, the abbreviation QL is used without explanation. Its full form and meaning should be clearly defined.
The abbreviation CV should be expanded at its first occurrence in the text. All abbreviations should be introduced with their full forms when first used, followed by the abbreviated forms in subsequent occurrences.

Reviewer #6

Questions

2. The paper is relevant to the scope of AI in Healthcare and the selected conference track.
Strongly Disagree

3. The research methodology is appropriate, technically sound, and adequately explained.
Agree

4. The experimental results or evaluation adequately support the claims made in the paper.
Neutral

5. The paper discusses relevant prior work and provides appropriate references.
Agree

6. The paper is clearly written, logically organized, and understandable.
Agree

7. Describe the main strengths and contributions of the paper.
The paper proposes an Adaptive Hybrid Quantum-Classical Neural Network (Adaptive HQNN) for medical diagnosis, with the primary objective of reducing estimated quantum computation during inference by dynamically selecting the quantum-processing depth according to the confidence of

a preliminary classical prediction. The approach is evaluated on three publicly available medical classification datasets: Wisconsin Diagnostic Breast Cancer (WDBC), UCI Cleveland Heart Disease, and Pima Indians Diabetes.

One of the main strengths of the paper is its adaptive inference architecture. Unlike a fixed-depth hybrid quantum-classical neural network that executes the same quantum circuit for every sample, the proposed approach provides three inference paths: a classical-only path with zero quantum layers, a path with one quantum layer, and a path with two quantum layers. This design offers a practical strategy for allocating computational resources according to the model's preliminary prediction confidence.

A second strength is the evaluation across three medical datasets representing different diagnostic tasks and feature dimensionalities. The authors compare four models: a classical multilayer perceptron, a fixed-depth hybrid quantum-classical neural network, the proposed Adaptive HQNN, and an adaptive classical control model. This comparison is useful because it examines both the predictive-performance trade-off and whether adaptive computation can provide similar benefits without quantum layers.

The paper also demonstrates awareness of data leakage and evaluation reliability. The authors use an 80:20 development-test split, five-fold stratified cross-validation, training-only preprocessing statistics, and an inner cross-validation procedure for selecting routing thresholds. These measures strengthen the experimental design by separating threshold selection and preprocessing from the corresponding evaluation data.

The reported results indicate substantial reductions in estimated inference-time quantum-layer usage. The paper reports reductions of 98.8% for WDBC, 87.8% for Heart Disease, and 79.6% for Diabetes relative to the fixed two-layer HQNN. The WDBC results are particularly notable, with 98.0% of samples reportedly routed through the classical-only path. These findings support the paper's central argument that confidence-based routing can reduce the number of quantum layers used during inference.

Another positive aspect is the inclusion of multiple evaluation metrics, including accuracy, F1-score, ROC-AUC, and workload estimates. The authors also apply paired Wilcoxon signed-rank tests to compare ROC-AUC results across cross-validation folds and appropriately acknowledge that the absence of statistical significance does not establish equivalence between the models.

Furthermore, the paper incorporates Kernel SHAP to examine feature contributions to predictions and investigate the relationship between clinical feature distributions and routing decisions. The authors appropriately distinguish model explanations from causal clinical relationships. They also explicitly acknowledge that estimated quantum-layer reductions are not direct measurements of energy consumption, execution time, or power usage and do not demonstrate quantum advantage or clinical readiness.

Overall, the paper's main contributions are its adaptive quantum-layer routing architecture, leakage-conscious threshold-selection procedure, comparative evaluation across three medical classification datasets, quantitative analysis of estimated quantum workload, and explainability

analysis. The study offers a useful investigation of resource-aware inference in hybrid quantum-classical medical classification, while recognizing the trade-off between computational workload and predictive performance.

8. Identify the main weaknesses, limitations, and areas for improvement.
Although the paper presents a relevant approach to resource-aware medical classification, several aspects of its methodology, evaluation, and contribution require further clarification or improvement.

First, the originality of the proposed approach could be established more convincingly. Confidence-based routing and adaptive computation are established concepts in classical machine learning. The paper extends these ideas to the selection of quantum-layer depth, but the distinction between the proposed method and existing early-exit or conditional-computation architectures is not sufficiently quantified. A clearer comparison with existing adaptive inference strategies would help demonstrate the specific methodological contribution.

Second, the reported reduction in quantum workload is based on the average number of quantum layers executed per sample rather than direct measurements of execution time, energy consumption, or hardware resource utilization. Although the authors acknowledge this limitation, quantum-layer count alone may not accurately represent actual computational savings because circuit execution costs can depend on circuit structure, measurement requirements, simulation overhead, and hardware characteristics. The authors should validate the proposed efficiency claims through circuit-level resource estimates and, where feasible, execution-time measurements on quantum simulators or physical quantum processors.

Third, the predictive-performance results reveal a consistent trade-off that deserves deeper investigation. The Adaptive HQNN has lower cross-validation F1-scores than the fixed-depth HQNN on all three datasets. For example, on WDBC, the cross-validation F1-score decreases from 97.61% for the fixed HQNN to 95.88% for the Adaptive HQNN. On the Diabetes dataset, it decreases from 70.36% to 66.27%. These differences raise questions about whether the workload reduction justifies the loss in predictive performance for medical classification tasks. The paper would benefit from a more detailed analysis of the operating trade-off and criteria for selecting acceptable performance thresholds.

Fourth, the routing-threshold selection procedure requires additional detail. Although the authors describe an inner cross-validation process intended to prevent leakage, the precise threshold-selection objective, candidate threshold ranges, optimization criterion, and treatment of competing objectives should be explained more explicitly. The paper should also clarify how the preliminary classifier, quantum-path output heads, and routing thresholds are trained and selected in relation to one another. This would improve reproducibility and make it easier to determine whether the routing policy is being evaluated independently of model-selection decisions.

Fifth, the evaluation uses relatively small tabular datasets, particularly the Cleveland Heart Disease dataset with 303 observations. Although these datasets are established benchmarks, they do not represent the full complexity of clinical practice, where data distributions, patient populations, diagnostic procedures, and missing-data patterns may differ. Further evaluation on larger and more diverse datasets, followed by external validation, would be necessary before

making stronger claims about medical applicability.

Sixth, the statistical analysis has limited power because the paired Wilcoxon signed-rank tests are conducted on only five cross-validation folds. The authors appropriately acknowledge this limitation, but the resulting p-values cannot establish equivalence between the adaptive and fixed models. Repeated stratified cross-validation, additional independent test sets, and more robust uncertainty estimates would strengthen the comparison. The authors should also avoid interpreting small differences in point estimates as definitive evidence of superiority.

Seventh, the paper includes an adaptive classical control model, which is a useful baseline, but the implications of this comparison could be developed further. Because adaptive computation can reduce processing requirements even without quantum layers, the authors should clarify the extent to which the proposed benefit arises from the routing strategy itself and whether the quantum components offer any measurable advantage over an appropriately matched classical alternative. This is particularly important given that the paper does not claim quantum advantage.

Eighth, the explainability analysis could be strengthened. Kernel SHAP is used to identify features associated with predictions and routing decisions, but the paper provides limited quantitative detail about the stability of feature rankings across folds or models. Additional analysis of explanation consistency would improve confidence in the reported interpretations. The authors should continue to distinguish associations learned by the model from causal medical relationships.

Finally, further implementation details would improve reproducibility. These include the complete architecture and parameter counts, the exact threshold-selection objective, hyperparameter-selection procedures, quantum circuit gate counts, and details of the final holdout evaluation. Providing executable code and a clear configuration of the experimental pipeline would facilitate independent verification.

In summary, the most important areas for improvement are establishing the novelty of the adaptive routing mechanism, validating workload reductions beyond quantum-layer counts, clarifying threshold optimization, strengthening statistical evaluation, and demonstrating whether the efficiency gains justify the reduction in predictive performance. These improvements would help establish the practical significance of the proposed framework for resource-aware medical classification.

Reviewer #7

Questions

2. The paper is relevant to the scope of AI in Healthcare and the selected conference track.
Strongly Agree

3. The research methodology is appropriate, technically sound, and adequately explained.
Disagree

4. The experimental results or evaluation adequately support the claims made in the paper.
Disagree

5. The paper discusses relevant prior work and provides appropriate references.
Neutral

6. The paper is clearly written, logically organized, and understandable.
Agree

7. Describe the main strengths and contributions of the paper.
In my expert view, the manuscript investigates confidence-based adaptive inference in hybrid quantum-classical machine learning, proposing a three-path routing mechanism that dynamically allocates zero, one, or two quantum layers based on prediction confidence. The evaluation across three public medical classification datasets and comparisons with classical and fixed-depth quantum models provide an initial assessment of the proposed approach. Strengths include a clear architectural formulation, appropriate cross-validation and data-leakage prevention measures, and transparent reporting of predictive performance and estimated quantum-layer usage.

However, the adaptive approach does not consistently outperform the fixed-depth HQNN, and reduced quantum-layer usage alone does not establish quantum advantage or practical clinical applicability. The authors appropriately acknowledge these limitations. Overall, the manuscript presents a relevant preliminary investigation and provides a useful foundation for further research into resource-efficient hybrid quantum-classical inference.

8. Identify the main weaknesses, limitations, and areas for improvement.
In my humble expert opinion, the manuscript addresses a relevant & timely problem in hybrid quantum-classical machine learning by proposing confidence-based adaptive quantum-layer allocation. However, its technical novelty remains insufficiently established compared with existing classical conditional-computation methods. Further details on threshold selection, quantum-circuit design, training procedures, and the adaptive classical baseline are needed to improve reproducibility and ensure fair comparisons. Moreover, reduced quantum-layer usage does not necessarily translate into lower inference latency or energy consumption, particularly when experiments rely on a quantum simulator.

The evaluation would benefit from repeated experiments, uncertainty estimates, additional ablation studies, and clearer explanations of differences between cross-validation and holdout-test results. The limited medical datasets also restrict conclusions about clinical generalizability, while the related-work discussion and SHAP analysis require further clarification. Overall, the study offers a useful preliminary investigation, but stronger evidence of novelty, statistical reliability, reproducibility, and practical efficiency is needed to support its broader claims.

Reviewer #8

Questions

2. The paper is relevant to the scope of AI in Healthcare and the selected conference track.
Agree

3. The research methodology is appropriate, technically sound, and adequately explained.
Agree

4. The experimental results or evaluation adequately support the claims made in the paper.

Agree

5. The paper discusses relevant prior work and provides appropriate references.
Agree

6. The paper is clearly written, logically organized, and understandable.
Strongly Agree

7. Describe the main strengths and contributions of the paper.
A major strength is the evaluation across three medical classification datasets: Wisconsin Diagnostic Breast Cancer, Cleveland Heart Disease, and Pima Indians Diabetes. The inclusion of a classical multilayer perceptron, a fixed-depth hybrid quantum-classical network, and an adaptive classical control model provides useful comparative perspectives. The experimental methodology incorporates stratified cross-validation, training-partition-based preprocessing, nested inner cross-validation for routing-threshold selection, and a separate holdout test set.

8. Identify the main weaknesses, limitations, and areas for improvement.
The evaluation is limited to three relatively small, publicly available tabular datasets. Broader validation using larger, more diverse, and clinically representative datasets would help establish generalizability. External validation and more detailed analysis of sensitivity, specificity, false-negative rates, and probability calibration are also important before considering clinical applications.

Reviewer #9

Questions

2. The paper is relevant to the scope of AI in Healthcare and the selected conference track.
Strongly Disagree

3. The research methodology is appropriate, technically sound, and adequately explained.
Strongly Disagree

4. The experimental results or evaluation adequately support the claims made in the paper.
Strongly Disagree

5. The paper discusses relevant prior work and provides appropriate references.
Strongly Disagree

6. The paper is clearly written, logically organized, and understandable.
Strongly Disagree

7. Describe the main strengths and contributions of the paper.
• Resource-aware focus: Targets immediate computational constraints of NISQ computing platforms.
• Honest reporting: Explicitly maps out the performance-efficiency trade-off without exaggeration.
• Methodological hygiene: Implements robust inner cross-validation safeguards to rule out optimistic bias.
• Thorough baselines: Compares results directly against an Adaptive Classical Control model to isolate the actual cause of efficiency gains.
• Explainability integration: Employs global and routing-focused SHAP analyses to decipher why certain samples trigger deep quantum paths.

8. Identify the main weaknesses, limitations, and areas for improvement.

• Classical bottleneck: The routing engine depends entirely on the confidence of the preliminary classical head, making it susceptible to overconfident misclassifications.
• Clinical danger: Performance degrades across all datasets (e.g., F1 drops from 97.61% to 95.88% on WDBC), which is highly problematic for medical application deployments without override contingencies.
• Simulation limits: All execution reductions are purely software proxies; no real hardware deployment or noise profile evaluations were provided.

Reviewer #10

Questions

2. The paper is relevant to the scope of AI in Healthcare and the selected conference track.
Strongly Disagree

3. The research methodology is appropriate, technically sound, and adequately explained.
Agree

4. The experimental results or evaluation adequately support the claims made in the paper.
Agree

5. The paper discusses relevant prior work and provides appropriate references.
Agree

6. The paper is clearly written, logically organized, and understandable.
Agree

7. Describe the main strengths and contributions of the paper.
The manuscript presents an interesting and practical approach to improving the computational efficiency of hybrid quantum-classical neural networks for medical diagnosis. Its main contribution is an adaptive architecture that dynamically selects zero, one, or two quantum layers based on prediction confidence, rather than applying the same quantum computation to every sample.
A key strength is the evaluation across three medical datasets covering breast cancer, heart disease, and diabetes. The comparison with classical neural networks, fixed-depth hybrid models, and an adaptive classical baseline provides a useful perspective on the trade-off between predictive performance and computational workload.
I also appreciate the authors' efforts to reduce data leakage through nested cross-validation, include SHAP-based explanations, and acknowledge the limitations of quantum simulation. The reported reductions in estimated quantum-layer usage demonstrate the potential of adaptive routing for resource-aware AI systems.
Overall, the paper offers a meaningful contribution to hybrid quantum-classical machine learning, particularly through its emphasis on computational efficiency, transparent evaluation, and responsible interpretation of experimental results.

8. Identify the main weaknesses, limitations, and areas for improvement.
Although the proposed framework is promising, several limitations should be addressed to strengthen the contribution.
First, the experiments are conducted entirely using a quantum simulator rather than physical quantum hardware. Consequently, the reported workload reductions represent fewer executed quantum layers, not verified improvements in execution time, energy consumption, or hardware efficiency. The authors should provide runtime measurements and discuss the overhead

introduced by confidence-based routing.
Second, the adaptive model generally achieves lower cross-validation accuracy and F1-scores than the fixed-depth hybrid network. A more detailed analysis is needed to explain when the computational savings justify the reduction in predictive performance.
Third, the evaluation relies on three relatively small, publicly available medical datasets. Additional experiments using larger and more diverse clinical datasets, repeated random seeds, and independent validation would strengthen the generalizability of the findings.
Fourth, the novelty of the adaptive routing mechanism should be more clearly distinguished from established early-exit and dynamic neural network approaches. An ablation study examining routing thresholds, confidence calibration, and quantum-layer configurations would help establish the contribution of each component.
Finally, the statistical analysis is based on only five paired cross-validation folds, limiting the power of the significance tests. More comprehensive statistical evaluation, along with clearer explanations of the quantum circuit training procedure and SHAP results, would improve reproducibility.
These improvements would strengthen the technical evidence while preserving the paper's valuable contribution to resource-aware quantum-classical AI.

Reviewer #11

Questions

2. The paper is relevant to the scope of AI in Healthcare and the selected conference track.
Strongly Agree

3. The research methodology is appropriate, technically sound, and adequately explained.
Neutral

4. The experimental results or evaluation adequately support the claims made in the paper.
Neutral

5. The paper discusses relevant prior work and provides appropriate references.
Agree

6. The paper is clearly written, logically organized, and understandable.
Agree

7. Describe the main strengths and contributions of the paper.
The paper addresses an interesting and relevant problem: reducing unnecessary quantum computation during inference in hybrid quantum-classical neural networks for medical classification. The proposed Adaptive HQNN introduces a confidence-based routing mechanism that selects among classical-only, one-quantum-layer, and two-quantum-layer inference paths. This is a straightforward and potentially useful extension of adaptive computation to hybrid quantum-classical models.

A major strength is the explicit consideration of the trade-off between predictive performance and estimated quantum workload. The evaluation across three public medical datasets—Wisconsin Diagnostic Breast Cancer, Cleveland Heart Disease, and Pima Indians Diabetes—provides some evidence of how the routing strategy behaves across different binary classification tasks. The reported reductions in average quantum-layer usage are substantial, ranging from 79.6% to

98.8% relative to the fixed two-layer HQNN baseline.

The inclusion of both a conventional classical MLP and an adaptive classical control model is also valuable, as it helps distinguish the benefits of conditional computation from those attributable specifically to quantum components. The paper further describes a nested threshold-selection procedure intended to prevent information leakage, along with training-fold-specific preprocessing and an isolated holdout test set.

Another positive aspect is the authors' cautious interpretation of their findings. They acknowledge that reduced quantum-layer usage is only a proxy for computational savings and do not claim quantum advantage or clinical readiness. The SHAP-based analysis adds an interpretability perspective by examining predictive features and their associations with routing decisions.

Overall, the paper presents a clearly motivated resource-allocation strategy, a relevant experimental comparison, and a useful discussion of the limitations of the current evidence.

8. Identify the main weaknesses, limitations, and areas for improvement.

The primary limitation is that the novelty and significance of the contribution remain somewhat limited. Confidence-based adaptive computation and early-exit mechanisms are well established in classical machine learning, and the proposed method largely transfers this principle to a hybrid quantum-classical architecture. The paper would benefit from a clearer explanation of what is technically distinctive about adaptive quantum-layer selection beyond applying an existing routing concept to quantum circuits.

A second concern is the limited evidence for practical computational benefits. The workload metric counts the average number of quantum layers executed per sample but does not measure actual inference latency, circuit evaluations, simulator runtime, energy consumption, or hardware resource utilization. Moreover, the experiments use the default.qubit simulator rather than physical quantum hardware. Consequently, the substantial reductions in layer usage do not yet establish corresponding reductions in real-world inference cost.

The predictive results also require more rigorous interpretation. The Adaptive HQNN has lower cross-validation F1 scores than the fixed-depth HQNN on all three datasets, and its performance does not consistently exceed that of the classical MLP. Although the paper reports paired Wilcoxon signed-rank tests, five cross-validation folds provide limited statistical power. More robust repeated-seed experiments, uncertainty intervals, and appropriate statistical comparisons would strengthen the conclusions.

The threshold-selection objective and its implementation should be described in greater detail. In particular, the paper should clarify the candidate threshold ranges, optimization objective, handling of class imbalance, and whether the routing thresholds and final model parameters are selected without any information from the holdout test set. The interaction between the preliminary classifier, routing decisions, and final predictions also deserves a more detailed analysis.

The experimental scope is limited to three relatively small, established tabular datasets. These benchmarks do not establish generalizability to larger, more diverse, or clinically representative populations. The paper should also provide more detailed implementation information, including

the number of trainable parameters, circuit-evaluation budget, measurement assumptions, and reproducibility details.

Finally, the Adaptive Classical Control baseline should be analyzed more deeply to determine whether a comparable reduction in computation can be obtained through a purely classical architecture with similar predictive performance. A comparison against simpler uncertainty-based selective prediction or early-exit baselines would help establish the specific value of the proposed quantum routing mechanism.

In summary, the paper presents a promising exploratory idea, but stronger methodological transparency, more comprehensive baselines, direct resource measurements, and more robust evaluation are needed before its practical significance can be established.